



OPERATIONAL TECHNOLOGY (OT) CYBERSECURITY TRAINING

FACILITATOR GUIDE

FACILITATOR INSTRUCTIONS (READ FIRST)

You do **not** need to be a cybersecurity expert to lead this training.

Your role is to walk through the material, explain the examples, reinforce safe behaviors, and encourage discussion.

If someone asks a question you cannot answer, say: "That's a great question. I'll bring it to our cybersecurity team so you get the correct information."

Keep the session practical and conversational. Discussions about abnormal behavior, near misses, and real-world examples are extremely valuable.

TRAINING SCRIPT (FACILITATOR READS ALOUD)

1. Introduction: What is OT?

Operational Technology (OT) is a specific category of systems defined in regulation as:

Programmable systems or devices that interact with the physical environment (or manage devices that interact with the physical environment). These systems or devices detect or cause a change through the monitoring or control of devices, processes, and events.

Put simply:

- OT includes any system that directly controls physical equipment or monitors conditions that cause equipment to move or respond.
- OT is different from IT (email, office computers, business systems), because OT affects real equipment, real processes, and real safety.

2. Examples of OT Systems

Below are examples of OT systems commonly found at facilities and vessels:

- Pumps, valves, motors, actuators (devices that cause movement or change)
- PLCs (Programmable Logic Controllers) that issue commands to equipment
- HMIs (Human-Machine Interfaces) or operator screens used to monitor and control operations
- SCADA systems that collect data and control processes remotely
- Alarm and safety shutdown systems that detect conditions and trigger responses
- Power distribution and automated breaker systems
- Cargo handling, transfer, and loading automation
- Navigation or propulsion control systems that control or influence vessel movement (AIS, ECDIS, autopilot)



These all meet the definition because they interact with or directly manage the physical environment.

3. Why OT Cybersecurity Matters

Cyberattacks aren't just about stealing data. When OT systems are affected, the impact can be physical, not just digital. Cyber issues in OT can lead to:

- Equipment operating in unexpected or unsafe ways
- Loss of visibility or control
- Safety or environmental hazards
- Operational disruption

What this can look like in real life:

- A pump turning on or off unexpectedly
- A valve opening or closing without a command
- Alarms or operator screens showing incorrect information
- Sensors providing inconsistent or impossible values
- Propulsion, steering, or thruster responding differently than commanded
- Navigation or bridge systems (AIS, ECDIS, autopilot) showing conflicting or unreliable data

4. Common OT Cyber Risks and How They Occur

Most OT cybersecurity problems start with everyday actions, not hacking. The goal here is to understand what kinds of behaviors create risk, so you know what to avoid.

Here are common ways OT systems can be put at risk, whether intentionally or accidentally:

Access & Credentials

- Using someone else's login or sharing your own
- Skipping login steps to save time

Devices & Connections

- Plugging in personal or unknown removable hardware (e.g., USB drives)
- Allowing a contractor or vendor to connect equipment without approval
- Connecting personal devices or unapproved tools

Physical Security

- Propping open secure OT-area doors
- Letting someone into an OT space without verifying who they are

System Behavior

- Strange pop-ups, alarms, unexpected screens, or error messages on OT computers
- Phone calls or emails asking you to "reset" equipment or "check a reading"
- Changing setpoints or controls outside of procedure



These examples illustrate how small actions—often done to "get the job done"—can create big cybersecurity and safety risks. If the procedure feels like it slows you down, it's because it protects operations.

If something feels unusual or you're unsure whether something is allowed, stop and ask.

5. How to Recognize When Something Is Wrong

You don't have to diagnose the problem. You only need to recognize unusual behavior. Signs you might be seeing a cyber-related issue include:

- Equipment behaves unexpectedly or in a way you didn't command
- A Human-Machine Interface (HMI) or operator screen freezes, flickers, or shows inconsistent data
- You lose communication with a device
- Login doesn't work even though you're entering it correctly
- New alarms appear without explanation
- Unknown USB devices or cables are connected
- A vendor asks to connect equipment you've never seen before
- Setpoints changing on their own

When in doubt, treat unexpected behavior as suspicious and report it immediately.

6. What To Do If You Notice Something

If you see something unusual, report it right away. You will never get in trouble for reporting a concern that turns out to be nothing.

Report to:

1. Supervisor / Operations Lead
2. Facility / Vessel / Company Security Officer (FSO / VSO / CSO)
3. IT ServiceDesk
4. Cybersecurity Officer (CySO)

Note for facilitator: *If asked whether to shut something down, say, "Follow safety procedures. For cybersecurity questions, contact the CySO."*

7. Third-Party and Remote Access Rules

Vendors, contractors, and other third parties must follow strict rules when accessing OT systems. Their level of access depends on whether they have completed the required cybersecurity training.

Trained vendors may be authorized for:

- Certain types of unescorted access (as approved by the company)
- Performing maintenance or troubleshooting with less direct oversight
- Remote access sessions that follow approved procedures and monitoring requirements



Untrained vendors may **only** access OT systems if they are:

- Physically accompanied by trained personnel at all times, or
- Remotely monitored by trained personnel using approved escorting methods

You should never allow any vendor—trained or untrained—to:

- Plug into systems or equipment without authorization
- Access systems remotely without prior approval and scheduling
- Perform work outside the scope of what has been authorized

If you aren't sure, stop and ask or say, "I need to check with our cybersecurity team before we proceed."

8. Wrap-Up

Your awareness is one of the strongest cybersecurity defenses we have. You are responsible for:

- Protecting your devices and login credentials
- Keeping secure areas secure
- Reporting unusual behavior
- Using only approved tools and devices
- Following established procedures
- Asking questions rather than guessing

If something feels unusual, report it. If you're unsure, ask.

Facilitator questions for the group:

If you have time, use the discussion prompts below to open a dialogue with trainees:

- *What unusual OT behavior have you seen in the past?*
- *Who would you contact first if something seemed off?*

Remind employees they won't be penalized for reporting.